How does Module Array Design Affect the Food-Energy Productivity of Agrivoltaic Systems?

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***Abstract*— Agrivoltaics (AV) is an innovative approach for dual food-electricity production on the same land that preserves land ecosystems and can enable several food–energy–water synergies. The design of the module arrays for AV needs an optimal balance for sharing sunlight between the agricultural and energy production. We explore the module array design for various AV configurations using a basic approach to evaluate AV performance relative to the standard ground-mounted PV as a function of the module array density, elevation, spatial distribution of light and the crop yield for different crop sensitivities. A spatial-temporal irradiance sharing model between the solar modules and crops is used to estimate the food-energy yields. Case studies are presented for different Agrivoltaic (AV) orientations, including a) North/South fixed tilt, b) vertical East/West faced bifacial, and c) single-axis tracking, specifically solar tracking (ST) and anti-tracking (AT) systems. These configurations are evaluated with various crops representing different levels of shade sensitivities. The presented approach is applicable to any global location and can be used for module array design, crop selection, and policy decisions.**

Keywords—agrivoltaics, food-energy yield, module array design, spatial light distribution

I. INTRODUCTION

Agrivoltaics (AV) is a dual land-use technology that can allow a win-win solution for deploying large-scale photovoltaic installations over agricultural in terms of preserving the ecosystem and biodiversity while enabling several synergistic benefits to food-energy-water nexus [1]. Favorable microclimates in AV farms protect crop yields in heat/drought stress, enabling diversification of species with positive implications for nutrition, and shelter for livestock [2]. It also offers a reliable supply of electricity to feed the increasing energy demands of agriculture, e. g. irrigation, and can also service the needs of local food storage and processing, thereby saving a tremendous amount of food wastage. Moreover, reduced evapotranspiration under solar panels reduces irrigation needs thus protecting against water shortfall [2]. A dual source of revenue from crops and energy production can serve an important function of socio-economic uplift of the local farmers [3][4].

To make an AV system locally attractive, technological solutions need to be developed through which both energy and agriculture production can be maximized. Tracking solutions for AV, requires models that can address the distinctive challenges and considerations to cater for a broad range of crop shade response and food-energy yield requirements. Since the standard tracking (ST) is designed to maximize the energy yield, it can result in a relatively higher shading for crops while anti tracking prioritizes the food production by compromising on energy yield in a given AV module configuration. Therefore, careful selection of AV orientation and module array design can aid in maximizing food and energy yield.

Although many innovative AV approaches have been demonstrated across several locations in the world, comprehensive modeling that explores and optimizes food-energy production for a variety of module array configurations has not been reported. Here, we present our work on the AV farm productivity modeling and explore the design of efficient AV systems in terms of food-energy yield requirements for several bifacial module arrays focusing on N/S faced fixed tilt b) Vertical E/W faced c) solar tracking and d) anti tracking AV systems. The rest of the paper includes modeling approach in Section II, food-energy productivity, and effect of ground clearance (elevation) are described under the results and discussion in Section III. Finally, conclusions are presented in Section IV.

II. MODELING APPROACH

A. PV Energy and incident PAR over crops

We use a view factor-based approach which has been previously reported and validated on field experiments [5-7]. Briefly, the model calculates sunlight interception by panels to get PV yield including the contributions from direct beam, diffused light and albedo. The shading for the direct beam and diffused light due to solar panels are evaluated to calculate the available PAR at any horizontal surface on the ground or at an elevation underneath the panel arrays. Typical meteorological conditions are used to estimate the diffused and direct components of the global horizontal irradiation [6]. An isotropic model is assumed for diffused light. Computations are carried out on 1-minute resolution. The analysis is done for $\boldsymbol{N/S}$ faced 30° fixed-tilt, $\boldsymbol{E/W}$ faced vertically faced bifacial, solar tracking ($\boldsymbol{ST}$) and anti-tracking ($\boldsymbol{AT}$) orientations as shown in Fig. 1. In standard tracking (ST) scheme, the modules track the sun prioritizing energy generation over food production, and in anti-tacking (AT) scheme, the face of the modules is kept parallel to the direct beam of sunlight prioritizing food production over energy generation.

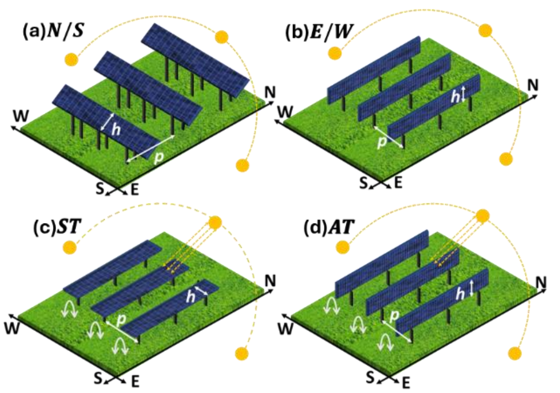


Fig. 1. Conceptual depiction of N/S fixed tilt, E/W vertical bifacial, ST and AT module systems with row-to row pitch(p) and module height(h) as design parameters.

B. Relative Crop-Energy Yields

The energy ratio (Y_{PV}) is the ratio of energy produced per unit module area of a given module configuration for AV to the energy produced per unit module area of the standard fixed tilt reference $GMPV$.

$$Y_{PV} = \frac{\textit{Annual energy produced per unit module area for AV}}{\textit{Annual energy converted per unit module area for GMPV}} \quad (1)$$

The sunlight availability to the crops is evaluated through the PAR(Photosythically active radiation) ratio (Y_{PAR}) which is a function of position (x) along the pitch between the module rows and time (t):

$$Y_{PAR}(x,t) = \frac{\textit{PAR available to crop for a given AV system}}{\textit{PAR available to crop for the reference full sun condition}} \quad (2)$$

C. Shade Sensitivities for Crop

Crop yield reduction resulting from shading is quantified by assessing the decrease in the PAR received by the crop across the crop cycle. Y_{Crop} is defined as the percentage biomass yield at the time of harvest for a crop under the AV shading relative to the reference full sun condition.

$$Y_{Crop} = f(Y_{PAR}) = \frac{\textit{Biomass yield for a crop in AV system}}{\textit{Biomass yield for the same crop in full sun condition}} \quad (3)$$

Fig. 2 shows the average response of Y_{crop} to PAR from the results reproduced from [8]. Crops having different shade sensitivities are classified as (i) shade susceptible (S) which are highly susceptible to shade, (ii) shade tolerant (T) which are moderately affected by shade, and (iii) shade benefiting (B) which are mildly affected by shade.

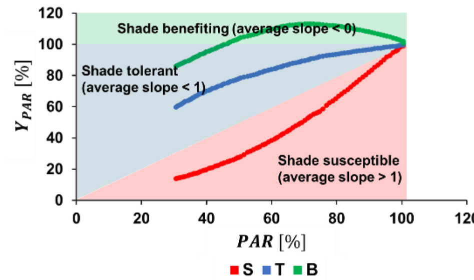


Fig. 2. Crop responses to shade with varying sensitivities, adapted from [8].

III. Results and Discussion

We use the location of Lahore (31.5204° N, 74.3587° E) for illustration throughout the simulations.

A. Food-energy yield for different module configurations and crops

Fig. 3 shows the impact of various crop shade sensitivities (crop types S, T, and B) on Y_{crop} and Y_{PV} for a range of p/h and using four different AV module configurations. Y_{crop} is computed using the data shown in Fig. 3 for the average annual Y_{PAR}. For crops type B, Y_{crop} is mildly affected by p/h and module configurations. For crop type S, Y_{Crop} is highly dependent on module configuration as well as on p/h increasing significantly with increase in p/h. At a constant p/h, AT is best suited for the shade sensitive crops, followed by E/W faced vertical and the N/S faced fixed tilt orientations. Energy yield (Y_{PV}) is highest for ST followed by N/S fixed tilt, E/W vertical and anti-tracking systems. Fig. 4 illustrates that Y_{crop} and Y_{PV} can have a broad range depending on the tracking scheme with ST and AT along the full day providing the upper and the lower limits.

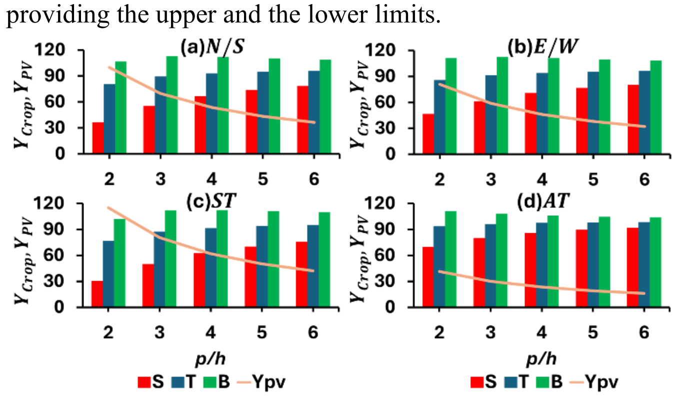
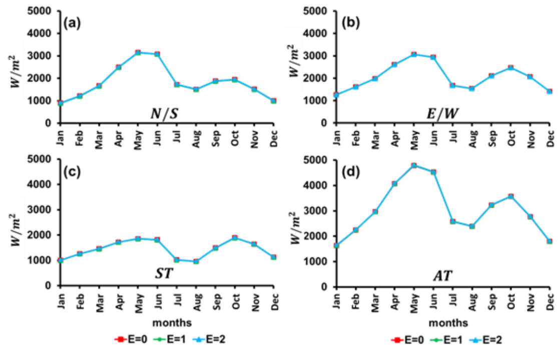


Fig. 3. Effect of different module configurations on Y_{Crop} and Y_{PV} for different crop sensitivities for range of p/h. Y_{Crop} increases with increasing p/h which is most evident in the crop type S while Y_{PV} decrease with increasing p/h. ST is not recommended for crop type S at $p/h \leq 4$.

B. Effect of module ground clearance (elevation above ground).

The impact of ground clearance or elevation of modules on the quantity and spatial distribution of sunlight reaching to the ground for different AV orientations is explored in Fig. 5 – 8.

Fig. 4 and 5 show the daily cumulative values averaged across the pitch for direct and diffused irradiation, respectively, on ground for different elevations across all months of the year while Fig. 6 and 7 show the spatial profiles of daily cumulative direct and diffused irradiation, respectively, along the pitch on the ground for various ground clearance (0, 1m, and 2m) across different months. It can be observed that the primary effect of varying the module elevation is on the spatial pattern of direct and diffused irradiation. This effect is most dominant when the elevation is increased from 0 to 1m. The spatially average values for different elevations are similar for direct light for all months. For diffused light, elevation of 0 results in a slightly higher spatially average values as compared to those at the higher elevations for various months across the year. From these results, we conclude that the module elevation is an important design parameter primarily for the spatial distribution of light on the ground.



Fig, 4. Daily cumulative direct irradiation on ground averaged across the pitch for months across the year at $p/h = 2$ for module elevation (ground clearance) of *0, 1m,* and *2m* above ground.

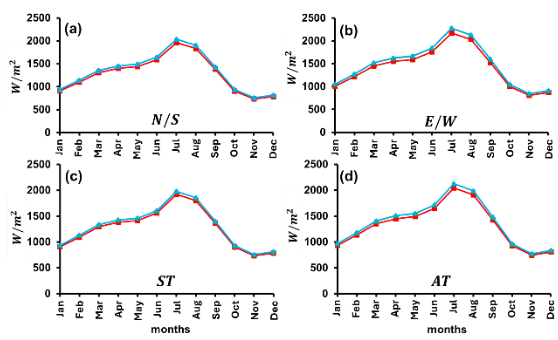


Fig. 5 Daily cumulative diffuse irradiation on ground averaged across the pitch for months across the year at $p/h = 2$ for module elevation (ground clearance) of *0, 1m,* and *2m* above ground.

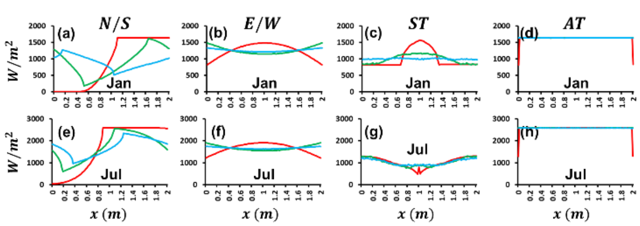


Fig. 6. Spatial distribution of daily cumulative direct irradiation on ground across the position along the pitch between the module rows at $p/h =\ 2$ for module elevation of $0, 1m$, and $2m$ above ground.

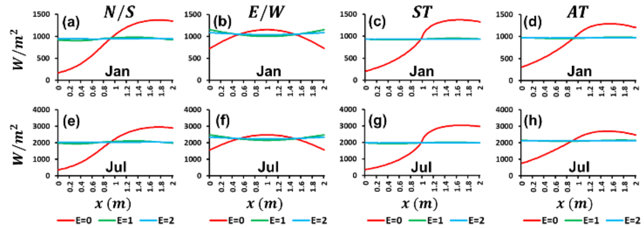


Fig. 7. Spatial distribution of daily cumulative diffuse irradiation on ground across the position along the pitch between the module rows at p/h= 2 for module elevation of $0, 1m$, and $2m$ above ground.

IV. Conclusions

Food-energy yield for various crop sensitivities for different orientations of agrivoltaics is evaluated using a spatial-temporal irradiation sharing model between the solar modules and the crops. Amongst different AV orientations, anti-tracking (AT) scheme can provide a higher crop yield but at a reduced energy generation while solar tracking (ST) scheme can provide highest energy generation by compromising on food yield at similar array density. Food-energy yield requirements for the shade benefitting crops can be supported with the standard sun tracking across the whole day for full (standard $p/h = 2$) module array density as well as with reduced module density. Food requirements for the crops with high shade sensitivity can either be achieved by reducing module density ($p/h > 4$) or through anti tracking. But energy yield requirements for AV are not met for AT so, shade sensitive crops for high module density ($p/h = 2$) are not feasible for AV systems. The impact of elevation on diffuse and direct irradiation along with the spatial distribution of light along the pitch is also evaluated which is dominant when elevation is increased from 0 to $1m$.

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